

25.3 Confounding and causal graphs

The functions and disturbances of a structural causal model are usually unknown; its parent sets, saying which variables can influence which, are qualitative and best displayed as a graph. This section shows how the graph constrains the joint distribution and identifies confounding as a property of paths. For linear models the connection is exact, and the omitted-variable bias of Proposition 5.7.2 becomes a sum over paths.

25.3.1 Graphs

Definition 25.3.1 — Directed acyclic graph

A **directed graph** on nodes $1, \dots, m$ is a set of ordered pairs (l, j) , $l \neq j$, drawn as arrows $l \rightarrow j$. It is **acyclic** (a DAG) if no sequence of arrows leads from a node back to itself. In a DAG:

- (a) l is a **parent** of j , and j a **child** of l , if $l \rightarrow j$;
- (b) a **path** between l and j is a sequence of distinct nodes, starting at l and ending at j , in which consecutive nodes are joined by an arrow in either direction;
- (c) a path is **directed** (from l to j) if every arrow on it points towards j ; then l is an **ancestor** of j , and j a **descendant** of l ;
- (d) an interior node c of a path is a **collider** on the path if both arrows of the path that meet at c point into it, $\rightarrow c \leftarrow$; every other interior node is a **non-collider** on the path.

The **graph of a structural causal model** has an arrow $l \rightarrow j$ iff $l \in \text{pa}(j)$. It is acyclic because parents have smaller indices.

Being a collider is a property of a node *on a given path*: in Figure 25.3.1(c), C is a collider on $X \rightarrow C \leftarrow Y$ but not on a path that continues along an arrow $C \rightarrow D$.

25.3.2 What the graph says about the distribution

Assume that for each j and each value $\mathbf{v}$ of the parents, the variable $f_j(\mathbf{v}, U_j)$ has a density $p_j(\cdot \mid \mathbf{v})$ with respect to Lebesgue or counting measure.

Proposition 25.3.1 — Markov factorization and truncated factorization

Under this assumption:

- (a) $(V_1, \dots, V_m)$ has density

$$p(v_1, \dots, v_m) = \prod_{j=1}^m p_j(v_j \mid v_{\text{pa}(j)}), \quad (25.3.1)$$

and $p_j(\cdot \mid v_{\text{pa}(j)})$ is the conditional density of V_j given $V_1, \dots, V_{j-1}$, which depends on them only through the parents;

- (b) under $\text{do}(V_k = x)$, the other variables have density $\prod_{j \neq k} p_j(v_j \mid v_{\text{pa}(j)})$, evaluated with $v_k = x$.

Proof.

- (a) The variables $V_1, \dots, V_{j-1}$ are functions of $U_1, \dots, U_{j-1}$, which are independent of U_j . For independent inputs, the conditional distribution of $f_j(V_{\text{pa}(j)}, U_j)$ given $(V_1, \dots, V_{j-1}) = (v_1, \dots, v_{j-1})$ is the distribution of $f_j(v_{\text{pa}(j)}, U_j)$ (Billingsley 1995), whose density is $p_j(\cdot \mid v_{\text{pa}(j)})$. The factorization follows by multiplying these conditional densities in the order $j = 1, \dots, m$.
- (b) The modified model is a structural causal model for the variables other than V_k , in which V_k is the constant x and every other assignment and disturbance is unchanged. Apply (a) to it. ■

So an intervention deletes one factor and keeps the others, which are observational conditional densities; with every variable observed, the interventional distribution is computable from the observational one.

The linear case has an explicit form. Write a linear model as $\mathbf{V} = \mathbf{c} + \mathbf{B}\mathbf{V} + \mathbf{U}$, where $\mathbf{B} = (b_{jl})$ has $b_{jl} \neq 0$ only if $l \in \text{pa}(j)$, so $\mathbf{B}$ is strictly lower triangular, and $\mathbf{\Omega} = \text{Cov}(\mathbf{U}) = \text{diag}(\omega_1, \dots, \omega_m)$. The coefficient b_{jl} labels the arrow $l \rightarrow j$.

Proposition 25.3.2 — Path tracing

In a linear structural causal model:

- (a) $\mathbf{I} - \mathbf{B}$ is invertible, $\mathbf{V} = \mathbf{A}(\mathbf{c} + \mathbf{U})$ with $\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}$, and $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{\Omega}\mathbf{A}^\top$;
- (b) for $j \neq l$, the entry a_{jl} of $\mathbf{A}$ is the sum, over all directed paths from l to j , of the products of the coefficients on the arrows of the path (zero if there is none), and $a_{jj} = 1$;
- (c) $E(V_j \mid \text{do}(V_l = x)) = E(V_j \mid \text{do}(V_l = 0)) + a_{jl}x$.

The number a_{jl} is the **total effect** of V_l on V_j : the change in the interventional mean of V_j per unit change in the value imposed on V_l .

Proof.

- (a) $\mathbf{I} - \mathbf{B}$ is unit lower triangular, hence invertible, and $(\mathbf{I} - \mathbf{B})\mathbf{V} = \mathbf{c} + \mathbf{U}$. The covariance follows from Theorem 2.2.1.
- (b) $\mathbf{B}$ is strictly lower triangular, so $\mathbf{B}^m = \mathbf{0}$ and $(\mathbf{I} - \mathbf{B})(\mathbf{I} + \mathbf{B} + \dots + \mathbf{B}^{m-1}) = \mathbf{I} - \mathbf{B}^m = \mathbf{I}$. The (j, l) entry of $\mathbf{B}^r$ is $\sum b_{jk_1} b_{k_1 k_2} \dots b_{k_{r-1} l}$ over all $k_1, \dots, k_{r-1}$; a term is nonzero only if $l \rightarrow k_{r-1} \rightarrow \dots \rightarrow k_1 \rightarrow j$ are arrows, which is a directed path of length r (its nodes are distinct because the graph is acyclic). Summing over r gives (b).
- (c) The intervened model is linear with $\mathbf{B}$ replaced by $\tilde{\mathbf{B}}$, in which row l is zero, and

with $c_l + U_l$ replaced by x . A directed path out of l never uses an arrow into l , so the directed paths from l in the graph of $\tilde{\mathbf{B}}$ are those of $\mathbf{B}$, and by (b) the (j, l) entry of $(\mathbf{I} - \tilde{\mathbf{B}})^{-1}$ is a_{jl} . By (a) applied to the intervened model, V_j is $a_{jl}x$ plus terms not involving x .

■

This is Wright's method of path coefficients (Wright 1921, 1934); the covariance formula in (a) gives his rule for correlations ([Exercise \(C1\)](#)).

25.3.3 Chains, forks and colliders

Three configurations of three variables are the building blocks of everything the graph says about conditional independence ([Figure 25.3.1](#)).

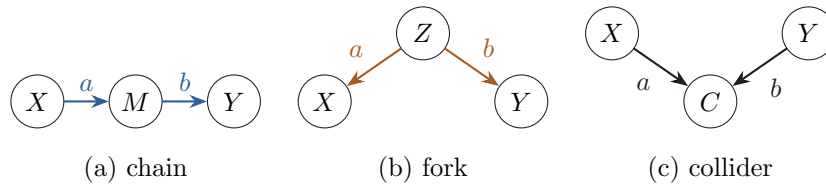


Figure 25.3.1. The three structures of [Proposition 25.3.3](#). In (a) and (b) the middle node transmits association between X and Y , and conditioning on it removes the association. In (c) the middle node transmits nothing until it is conditioned on.

Proposition 25.3.3 — Chains, forks and colliders

Consider linear models with independent disturbances, $\text{Var } U_X = \sigma_X^2 > 0$ and so on, and write $\sigma_{XY \cdot W}$ for the partial covariance of X and Y given W ([Definition 3.6.1](#)).

- (a) **Chain** $X = U_X$, $M = aX + U_M$, $Y = bM + U_Y$: $\text{Cov}(X, Y) = ab\sigma_X^2$ and $\sigma_{XY \cdot M} = 0$.
- (b) **Fork** $Z = U_Z$, $X = aZ + U_X$, $Y = bZ + U_Y$: $\text{Cov}(X, Y) = ab\sigma_Z^2$ and $\sigma_{XY \cdot Z} = 0$.
- (c) **Collider** $X = U_X$, $Y = U_Y$, $C = aX + bY + U_C$: $\text{Cov}(X, Y) = 0$ and

$$\sigma_{XY \cdot C} = -\frac{ab\sigma_X^2\sigma_Y^2}{a^2\sigma_X^2 + b^2\sigma_Y^2 + \sigma_C^2}. \quad (25.3.2)$$

With normal disturbances, a zero partial covariance means conditional independence ([Proposition 3.6.1](#)), so in (a) and (b) X and Y are dependent but conditionally independent given the middle node, and in (c) they are independent but, if $ab \neq 0$, conditionally dependent given C .

Proof. The covariances follow from [Proposition 25.3.2\(a\)](#). (a) $Y - bM = U_Y$ is uncorrelated with X and M , so $L(Y | M, X) = bM$ has no X term, and by [Proposition 14.5.2](#)

$\sigma_{XY \cdot M} = 0$. (b) The same argument with $Y - bZ = U_Y$. (c) By definition,

$$\sigma_{XY \cdot C} = \text{Cov}(X, Y) - \frac{\text{Cov}(X, C) \text{Cov}(Y, C)}{\text{Var}(C)} = 0 - \frac{(a\sigma_X^2)(b\sigma_Y^2)}{\text{Var}(C)}.$$

■

In the collider, independent causes become dependent once their common effect is held fixed. In the listing, with $a = b = 1$ and unit variances, the correlations of X and Y in the chain and the fork are 0.577 and 0.500 (simulated: 0.582 and 0.505), and both partial correlations given the middle node are zero. In the collider the correlation is zero and the partial correlation given C is -0.500 (simulated -0.501).

```
import numpy as np

rng = np.random.default_rng(2503)
n = 100_000
a, b = 1.0, 1.0                                # the two edge coefficients

def corr_and_partial(x, y, w):
    """Correlation of x and y, and their partial correlation given w (residuals on
    1, w)."""
    W = np.column_stack([np.ones(len(w)), w])
    rx = x - W @ np.linalg.lstsq(W, x, rcond=None)[0]
    ry = y - W @ np.linalg.lstsq(W, y, rcond=None)[0]
    return np.corrcoef(x, y)[0, 1], np.corrcoef(rx, ry)[0, 1]

U = rng.normal(size=(3, n))
x = U[0]; m = a * x + U[1]; y = b * m + U[2]      # chain    X -> M -> Y
chain = corr_and_partial(x, y, m)
z = U[0]; x = a * z + U[1]; y = b * z + U[2]      # fork      X <- Z -> Y
fork = corr_and_partial(x, y, z)
x = U[0]; y = U[1]; c = a * x + b * y + U[2]      # collider  X -> C <- Y
collider = corr_and_partial(x, y, c)
for name, (r, rp) in [("chain", chain), ("fork", fork), ("collider", collider)]:
    print(f"{name:9s} corr {r:6.3f}    partial corr given middle {rp:6.3f}")
```

[Run this code online](#)

25.3.4 d-separation

In a larger graph association can travel along many paths at once; d-separation combines the three structures.

Definition 25.3.2 — d-separation

Let $\mathcal{Z}$ be a set of nodes. A path is **blocked** by $\mathcal{Z}$ if

- (i) some non-collider on the path is in $\mathcal{Z}$, or
- (ii) some collider on the path is not in $\mathcal{Z}$ and has no descendant in $\mathcal{Z}$.

Otherwise the path is **open** given $\mathcal{Z}$. Disjoint sets of nodes $\mathcal{A}$ and $\mathcal{B}$ are **d-separated**

by $\mathcal{Z}$ (disjoint from both) if every path between a node of $\mathcal{A}$ and a node of $\mathcal{B}$ is blocked by $\mathcal{Z}$.

A path is open when each non-collider is left free and each collider is conditioned on, directly or through a descendant, a noisy measurement of it ([Exercise \(B1\)](#)).

Theorem 25.3.4 — d-separation and conditional independence

Let $\mathbf{V}$ follow a structural causal model with graph G , and let $\mathcal{A}, \mathcal{B}, \mathcal{Z}$ be disjoint sets of nodes.

- (a) If $\mathcal{Z}$ d-separates $\mathcal{A}$ and $\mathcal{B}$ in G , then $\mathbf{V}_{\mathcal{A}}$ and $\mathbf{V}_{\mathcal{B}}$ are conditionally independent given $\mathbf{V}_{\mathcal{Z}}$.
- (b) If $\mathcal{Z}$ does not d-separate them, there is a linear structural causal model on G with normal disturbances in which they are conditionally dependent; in fact this holds for all edge coefficients and disturbance variances outside a set of Lebesgue measure zero.

Proof. Sketch; full proofs are in the references. For (a) with densities (Lauritzen, Dawid, Larsen and Leimer 1990): the nodes of $\mathcal{A} \cup \mathcal{B} \cup \mathcal{Z}$ and their ancestors form a set $\mathcal{N}$ whose density, by [Proposition 25.3.1](#), is a product of functions of the sets $\{j\} \cup \text{pa}(j)$, $j \in \mathcal{N}$. These sets are complete in the *moral graph* on $\mathcal{N}$, which joins each node to its parents and any two parents of a common child. If $\mathcal{Z}$ separates $\mathcal{A}$ from $\mathcal{B}$ there, the factors split into two groups, giving the conditional independence; a combinatorial lemma shows that d-separation in G is equivalent to this separation. Geiger, Verma and Pearl (1990) give another proof. For (b), partial correlations of a linear normal model are rational functions of the parameters, not identically zero along an open path, and so vanish only on a set of measure zero (Spirtes, Glymour and Scheines 2000). ■

Part (b) says the converse of (a) fails only by accident, as in [Exercise \(A1\)](#), where the effect of visits and the association through illness cancel exactly. Distributions with no such cancellation are called **faithful** to the graph.

25.3.5 Confounding as an open path

Definition 25.3.3 — Back-door paths and confounding

A **back-door path** from X to Y is a path between them whose arrow at X points into X , $X \leftarrow \cdots Y$. The effect of X on Y is **confounded** if some back-door path from X to Y is open given the empty set, that is, contains no collider.

Directed paths from X to Y carry the effect of X ; back-door paths carry association that reflects whatever caused X . In a linear model the split is exact.

Proposition 25.3.5 — Confounding bias in a linear model

In a linear structural causal model with disturbances of positive variance, let $X = V_k$, $Y = V_j$, and let $\tau = a_{jk}$ be the total effect of X on Y . For each l , let d_l be the sum over the directed paths from V_l to Y that do not pass through X of the products of coefficients (so $d_k = 0$, and $d_j = 1$ from the trivial path of Y to itself).

- (a) $Y = \tau X + R$, where $R = \sum_l d_l(c_l + U_l)$ is the value of Y under $\text{do}(X = 0)$. Under $\text{do}(X = x)$, $Y = \tau x + R$.
- (b) The slope of the best linear predictor $L(Y | X)$ is

$$\tau + \frac{\text{Cov}(R, X)}{\text{Var}(X)}, \quad \text{Cov}(R, X) = \sum_l \omega_l d_l a_{kl}. \quad (25.3.3)$$

- (c) If the effect of X on Y is not confounded, the slope of $L(Y | X)$ is τ .

Proof.

- (a) A directed path from V_l to Y either avoids X or passes through it exactly once, and a path of the second kind splits at X into a directed path from V_l to X and one from X to Y . Summing products over paths with Proposition 25.3.2(b), $a_{jl} = d_l + a_{kl}\tau$ for every l (for $l = k$, $a_{kk} = 1$ and $d_k = 0$; for $l = j$, $a_{jj} = 1 = d_j$ and $a_{kj} = 0$). Hence $Y = \sum_l a_{jl}(c_l + U_l) = \tau \sum_l a_{kl}(c_l + U_l) + R = \tau X + R$. Under $\text{do}(X = x)$, as in the proof of Proposition 25.3.2(c), the arrows into X are removed, so the directed paths from V_l , $l \neq k$, to Y in the intervened graph are exactly those that avoid X . Thus $Y = \tau x + \sum_{l \neq k} d_l(c_l + U_l) = \tau x + R$.
- (b) The slope is $\text{Cov}(X, Y) / \text{Var}(X) = \tau + \text{Cov}(R, X) / \text{Var}(X)$ by Theorem 6.11.1, and $\text{Cov}(R, X) = \sum_{l, l'} d_l a_{kl'} \text{Cov}(U_l, U_{l'}) = \sum_l \omega_l d_l a_{kl}$, since the disturbances are uncorrelated.
- (c) Suppose $\omega_l d_l a_{kl} \neq 0$ for some l . Then $l \neq k$, and there are a directed path π_1 from V_l to X and a directed path π_2 from V_l to Y that avoids X . Let w be the node of π_1 closest to X that also lies on π_2 ; it exists because V_l lies on both, and $w \neq X$ because π_2 avoids X . Follow π_1 backwards from X to w , then π_2 forwards from w to Y . By the choice of w the nodes are distinct, so this is a path. Its arrow at X is the last arrow of π_1 , which points into X , and every arrow on it points away from w , so it has no collider. The effect is therefore confounded. Contrapositively, if it is not confounded, every term of $\text{Cov}(R, X)$ vanishes.

■

Part (a) is the potential outcome $Y(x) = \tau x + Y(0)$: linear structural models have constant effects. Part (b) writes the regression slope as the effect plus one contribution for each common ancestor of X and Y . In the model of Figure 25.1.1, $R = c_Y + \gamma Z + U_Y$ and the only term comes from Z : $\omega_Z a = \sigma_Z^2 \cdot \gamma \cdot \alpha$, which is Eq. 25.1.3.

This is the omitted-variable bias of least squares: by Proposition 5.7.2 the short regression estimates $\theta + \pi\gamma$, with π the sample slope of $\mathbf{z}$ on $\mathbf{x}$, whose limit is $\alpha\sigma_Z^2 / \text{Var}(X)$ (see also Theorem 19.1.2 in Chapter 19). The graph adds that an omitted variable causes bias only if it lies on an open back-door path; an omitted mediator should stay omitted (Section 25.5).

25.3.6 Exercises

25.3.6.1 A. Check your understanding

Exercise A1.

A DAG has arrows $A \rightarrow X$, $A \rightarrow B$, $B \rightarrow Y$, $X \rightarrow Y$, $X \rightarrow C$, $Y \rightarrow C$ and $D \rightarrow C$. List every path between X and Y , mark the back-door paths, and say which paths are open given $\emptyset$, given $\{B\}$ and given $\{C\}$.

Solution. There are three paths: $X \rightarrow Y$; $X \leftarrow A \rightarrow B \rightarrow Y$, a back-door path; and $X \rightarrow C \leftarrow Y$, with collider C . (The arrow $D \rightarrow C$ gives no path to Y except through C , where $D \rightarrow C \leftarrow Y$ would require revisiting C .) Given $\emptyset$: the first two are open and the third is blocked by the collider. Given $\{B\}$: only $X \rightarrow Y$ is open. Given $\{C\}$: all three are open, since conditioning on the collider opens the third.

25.3.6.2 B. Practice

Exercise B1.

Extend the collider by $D = C + U_D$, with $\text{Var } U_D = \delta$ and all other variances one. Compute $\sigma_{XY.D}$ and show that it lies between 0 and $\sigma_{XY.C}$, approaching them as $\delta \rightarrow \infty$ and $\delta \rightarrow 0$.

Solution. $\text{Cov}(X, D) = a$, $\text{Cov}(Y, D) = b$ and $\text{Var } D = a^2 + b^2 + 1 + \delta$, so $\sigma_{XY.D} = -ab/(a^2 + b^2 + 1 + \delta)$. At $\delta = 0$ this is [Eq. 25.3.2](#), and it decreases in absolute value to zero as δ grows. A proxy for a collider opens the path partially, the more so the better the proxy.

25.3.6.3 C. Going deeper

Exercise C1.

Show from [Proposition 25.3.2\(a\)](#) that in a linear model $\text{Cov}(V_j, V_k) = \sum_l \omega_l a_{jl} a_{kl}$. Interpret each term as a sum over pairs of directed paths from V_l to V_j and to V_k , and use this to compute $\text{Cov}(X, Y)$ in the chain of [Proposition 25.3.3](#).